A Subsethood Measure Based Approach for Handling Sparse Rule Bases Without Rule Interpolation

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***Abstract*—This paper introduces an analogical reasoning scheme that is grounded in lattice theory and uses a novel subsethood measure to handle sparse fuzzy rule bases by directly activating the most relevant rules. This methodology allows us to forego the use of rule interpolation. Based on this framework, an algorithmic strategy is proposed to address regression and time series prediction, constructing a particular fuzzy inference system for each training point and combining their outputs using a distance-weighted approach. Our paper includes several experimental results that indicate a competitive performance with fewer rules compared to interpolation-based methods. Future research will focus on optimizing parameter selection, improving computational efficiency, and enhancing robustness.**

***Index Terms*—subsethood measure, lattice theory, sparse rule base, fuzzy rule interpolation**

I. Introduction

The two most widely used fuzzy inference systems (FISs), namely Mamdani [1] and Takagi-Sugeno-Kang FISs [2], rely on a dense fuzzy rule base to generate meaningful outputs. However, in practice, this requirement is not always met. The available knowledge may be incomplete, or the need to reduce the complexity of the resulting FIS [3] may lead to a sparse rule base, increasing the risk of obtaining trivial outputs when the input does not overlap with any antecedent. To address this issue, fuzzy rule interpolation (FRI) techniques [4] have been extensively developed to infer conclusions by constructing new rules based on the input's relative position to existing antecedents. Various methods have been proposed [3], [5]–[8], demonstrating effectiveness in applications [9]–[11].

A less explored alternative for handling sparse rule bases involves using fuzzy measures for rule activation. For example, Kaburlasos and Kehagias utilized so-called "inclusion measures" [12]. An inclusion measure in the sense of Kaburlasos et al. [13] is a lattice-theoretical concept that is related to the one of a subsethood measure on a bounded lattice [14]. For our purposes, a certain bounded lattice of fuzzy numbers on an arbitrary non-empty, closed interval $[a, b]$ in $\mathbb{R}$ [15] is of utmost interest because this bounded lattice allows for defining a novel class of subsethood measures that can be used as matching functions in an analogical reasoning scheme (ARS) [16], [17]. We use the symbol $S^{\mathcal{A}}$ to denote an element of this new class of subsethood measures and refer to the resulting ARS as an $\mathcal{A}$-FIS. Generally speaking, an $\mathcal{A}$-FIS enables the activation of rules whose antecedents are closest to the input, even in non-overlapping conditions, making it well-suited for handling sparse rule bases. To enhance accuracy, we propose a complementary strategy that combines the distance between inputs and training points with optimized FISs for each training point.

The organization of this paper is as follows: Section II provides the necessary mathematical background on lattice theory and subsethood measures. Section III introduces the notion of an $S^{\mathcal{A}}$ subsethood measure that serves as a basis for the $\mathcal{A}$-FIS described in Section IV. We continue by comparing the results produced by the $\mathcal{A}$-FIS approach with the ones of two fuzzy rule interpolation methods [6], [18] in a number of benchmark regression problems. We finish by commenting on the key findings and future research directions.

II. Mathematical Preliminaries

A. Basic Concepts of Order and Lattice Theory

A *partially ordered set* or *poset* is a pair $(\mathbb{P}, \leq_{\mathbb{P}})$, where $\mathbb{P}$ is a non-empty set and $\leq_{\mathbb{P}}$ is a partial order relation on $\mathbb{P}$ [19], [20]. If the partial order relation arises clearly from the context, then we simply write $\leq$ instead of $\leq_{\mathbb{P}}$ and $\mathbb{P}$ instead of $(\mathbb{P}, \leq_{\mathbb{P}})$. A function $\varphi : \mathbb{P} \to \mathbb{P}'$, where $\mathbb{P}$ and $\mathbb{P}'$ are posets, is called *increasing*, if $x \leq y$ implies that $\varphi(x) \leq \varphi(y)$.

An element $l \in \mathbb{P}$ is called a *lower bound* of $X \subseteq \mathbb{P}$ if $l \leq x$ for all $x \in X$. If there is no lower bound m of X such that $l \leq m$, then l is called the greatest lower bound, *infimum*, or *meet* of X. The notion of a least upper bound, *supremum*, or *join* is defined similarly. The symbols $\bigwedge X$ and $\bigvee X$ denote respectively the infimum and the supremum of X. In the special case where $X = \{x, y\}$, one writes $x \wedge y$ and $x \vee y$. A poset $\mathbb{P}$ is said to be *bounded* if $\bigwedge \mathbb{P}$ and $\bigvee \mathbb{P}$ exist in $\mathbb{P}$, in which case these elements are denoted using the symbols $0_{\mathbb{P}}$ and $1_{\mathbb{P}}$, respectively. A poset $\mathbb{L}$ is called a *lattice* if we have for all $x, y \in \mathbb{L}$ that $x \wedge y$ and $x \vee y$ exist in $\mathbb{L}$. Recall that a lattice can also be defined as a non-empty set $\mathbb{L}$ together

with two idempotent, associative, and commutative operations $\vee, \wedge : \mathbb{L}^2 \to \mathbb{L}$ that satisfy the absorption laws [19], [20]. If these laws hold, then $\mathbb{L}$ is a lattice in which the partial order $\leq_{\mathbb{L}}$ is given by either one of the following statements:

- $x \leq_{\mathbb{L}} y$ if and only if $x \wedge y = x$.
- $x \leq_{\mathbb{L}} y$ if and only if $x \vee y = y$.

If $\mathbb{P}_1, \ldots, \mathbb{P}_n$ are posets, then a partial order on the direct product $\mathbb{P}_1 \times \ldots \times \mathbb{P}_n$ can be defined by setting for all $i \in \{1, \ldots, n\}$

$$(x_1, \ldots, x_n) \leq (y_1, \ldots, y_n) \Leftrightarrow x_i \leq y_i. \tag{1}$$

If $\mathbb{L}_1, \ldots, \mathbb{L}_n$ are bounded lattices, then $\mathbb{L}_1 \times \ldots \times \mathbb{L}_n$ is also a bounded lattice, called the *product lattice* with constituents $\mathbb{L}_1, \ldots, \mathbb{L}_n$. Given a non-empty set $\mathcal{U}$ and a poset $(\mathbb{L}, \leq)$, a partial order, denoted using the same symbol $\leq$, on the class of functions $\mathcal{U} \to \mathbb{L}$, denoted $\mathbb{L}^{\mathcal{U}}$, can be defined via

$$f \leq g \iff f(u) \leq g(u) \text{ , for all } u \in \mathcal{U}\,. \tag{2}$$

Recall that A is an $\mathbb{L}$-fuzzy set on $\mathcal{U} \neq \emptyset$ if and only if A is the graph of $\mu_A \in \mathbb{L}^{\mathcal{U}}$, called the membership function (MF) of A. For simplicity, we write $A(u)$ instead of $\mu_A(u)$ for $u \in \mathcal{U}$. The symbol $\mathcal{F}_{\mathbb{L}}(\mathcal{U})$ denotes the class of $\mathbb{L}$-fuzzy sets on $\mathcal{U}$. In the special case where $\mathbb{L} = [0,1]$, the symbol $\mathcal{F}(\mathcal{U})$ stands for $\mathcal{F}_{\mathbb{L}}(\mathcal{U})$. We have that $(\mathcal{F}(\mathcal{U}), \subseteq)$ is a bounded lattice. Here, the partial order $\subseteq$ is defined by $A \subseteq B$ if and only if $A(u) \leq B(u)$ for all $u \in \mathcal{U}$.

B. Fuzzy Subsethood Measures

Zadeh originally defined an *inclusion measure on* $\mathcal{F}(\mathcal{U})$ in terms of the aforementioned partial order relation $\subseteq$ on $\mathcal{F}(\mathcal{U})$ [21]. Several authors have axiomatically extended this concept to the one of *fuzzy set inclusion*, aka *fuzzy subsethood measure*, that quantifies the degree to which one fuzzy set (FS) is contained in another. Bustince et al. [22] thoroughly analyzed the relationships between these axiomatic definitions, demonstrating that Fan et al.'s definition of a subsethood measure [23] generalizes some important classical ones [24], [25]. An even more general definition is the following:

Definition 1. *Let $\mathbb{L}$ be a bounded poset. A function $S : \mathbb{L} \times \mathbb{L} \to [0,1]$ satisfying the following properties for all $x, y, z \in \mathbb{L}$ is said to be a* subsethood measure on $\mathbb{L}$.
SM1) If $x \leq_{\mathbb{L}} y$, then $S(x,y) = 1$;
SM2) $S(1_{\mathbb{L}}, 0_{\mathbb{L}}) = 0$;
SM3) If $x \leq_{\mathbb{L}} y \leq_{\mathbb{L}} z$, then $S(z,x) \leq S(y,x)$ and $S(z,x) \leq S(z,y)$. If $\mathbb{L} = [0,1]$, then we speak of a fuzzy subsethood measure.

Note that Definition 1 is a slight generalization of Sussner et al.'s definition [14] in which $\mathbb{L}$ is required to be a bounded lattice. A subsethood measure on a product of n bounded posets can be derived via an n-ary aggregation function from subsethood measures on its constituents. Recall the following definition [26]:

Definition 2. *Let $(\mathbb{P}, \leq, 0_{\mathbb{P}}, 1_{\mathbb{P}})$ be a bounded poset and $n \in \mathbb{N}$. A mapping $\oplus : \mathbb{P}^n \to \mathbb{P}$ is called an* (n-ary) aggregation function on $\mathbb{P}$ *if and only $\oplus$ satisfies the following conditions:*

1) $\oplus(\mathbf{x}) \leq \oplus(\mathbf{y})$ *whenever* $\mathbf{x} \leq \mathbf{y}$*;*
2) $\oplus(0_{\mathbb{P}}, \ldots, 0_{\mathbb{P}}) = 0_{\mathbb{P}}$ *and* $\oplus(1_{\mathbb{P}}, \ldots, 1_{\mathbb{P}}) = 1_{\mathbb{P}}$.

For simplicity, we write $\overset{n}{\underset{i=1}{\oplus}} x_i$ instead of $\oplus(x_1, \ldots, x_n)$.

Proposition 1. *Let $\mathbb{L} = \mathbb{L}_1 \times \cdots \times \mathbb{L}_n$, where $\mathbb{L}_i$ is a bounded poset for every $i \in \{1, \ldots, n\}$. Moreover, let $\oplus$ be an aggregation function on $[0,1]$. If $S_i : \mathbb{L}_i \times \mathbb{L}_i \to [0,1]$ is a subsethood measure for every $i \in \{1, \ldots, n\}$, then the function $S_{\oplus} : \mathbb{L} \times \mathbb{L} \to [0,1]$ given by Equation 3 is a fuzzy subsethood measure.*

$$S_{\oplus}(\mathbf{x}, \mathbf{y}) = \overset{n}{\underset{i=1}{\oplus}} S_i(x_i, y_i), \tag{3}$$
$$\text{for all } \mathbf{x} = (x_1, \ldots, x_n), \mathbf{y} = (y_1, ..., y_n) \in \mathbb{L}\,.$$

Proof. The proofs of SM1) and SM2) are straightforward. As for the proof for SM3), suppose that $\mathbf{x} \leq \mathbf{y} \leq \mathbf{z}$. Note that $x_i \leq y_i \leq z_i$ for all $i \in \{1, \ldots n\}$, which implies that $S_i(z_i, x_i) \leq S_i(y_i, x_i)$ and $S_i(z_i, x_i) \leq S_i(z_i, y_i)$. Since $\oplus$ is increasing,

$$S_{\oplus}(\mathbf{z}, \mathbf{x}) = \overset{n}{\underset{i=1}{\oplus}} S_i(z_i, x_i) \leq \overset{n}{\underset{i=1}{\oplus}} S_i(y_i, x_i) = S_{\oplus}(\mathbf{y}, \mathbf{x}), \tag{4}$$

$$S_{\oplus}(\mathbf{z}, \mathbf{x}) = \overset{n}{\underset{i=1}{\oplus}} S_i(z_i, x_i) \leq \overset{n}{\underset{i=1}{\oplus}} S_i(z_i, y_i) = S_{\oplus}(\mathbf{z}, \mathbf{y}). \tag{5}$$

□

A relevant construction method of subsethood measures, initially proposed for fuzzy sets [27] and later extended to any bounded lattice [14], involves the use of fuzzy implications.

Proposition 2. *Let $\mathbb{L}$ be a bounded lattice, $I : [0,1] \times [0,1] \to [0,1]$ be a fuzzy implication that satisfies $I(a,b) = 1$ for all $a \leq b \in [0,1]$, and $\nu : \mathbb{L} \to [0,1]$ be increasing such that $\nu(0_{\mathbb{L}}) = 0$ and $\nu(1_{\mathbb{L}}) = 1$. The following formula yields a subsethood measure on $\mathbb{L}$:*

$$S^{\vee}_{I,\nu}(x,y) = I(\nu(x \vee y), \nu(y)). \tag{6}$$

III. A NOVEL CLASS OF SUBSETHOOD MEASURES

In this section, we employ Proposition 2 in order to define a new class of subsethood measures on $\overline{\mathcal{RF}}(\mathcal{U})$, i.e., the so-called "bounded lattice of the fuzzy numbers on $\mathcal{U}$", where $\mathcal{U} = [v, w]$ is a closed, non-empty interval in $\mathbb{R}$. This bounded lattice was introduced in [15]. An element A of $\mathcal{F}(\mathcal{U})$ is said to be a *fuzzy number on $\mathcal{U}$* if we have for every $\alpha \in [0,1]$ that $[A]^{\alpha} = \{u \in \mathcal{U} \,|\, A(u) \geq \alpha\}$ is a closed, non-empty interval of the form $[\underline{a}^{\alpha}, \overline{a}^{\alpha}]$ in $\mathcal{U}$. The *class of fuzzy numbers on $\mathcal{U}$* is denoted using the symbol $\mathcal{RF}(\mathcal{U})$. Despite its name, which was chosen for convenience, the underlying set of the aforementioned bounded lattice is $\overline{\mathcal{RF}}(\mathcal{U}) = \mathcal{RF}(\mathcal{U}) \cup \{\emptyset\}$. The pairwise join in the bounded lattice $\overline{\mathcal{RF}}(\mathcal{U})$ is given by

$$(A \sqcup B)(u) = \begin{cases} 1, & \text{if } A, B \neq \emptyset, \text{ and} \\ & u \in [\underline{a}^1 \wedge \underline{b}^1, \overline{a}^1 \vee \overline{b}^1] \\ A(u) \vee B(u), & \text{otherwise} \end{cases} \tag{7}$$

The least and the greatest elements of $\overline{\mathcal{RF}}(\mathcal{U})$ are the fuzzy sets $\emptyset$ and $\mathcal{U}$ whose MFs are the constant functions 0 and 1, respectively.

We will use the binary operator $\sqcup$ of Equation 7 to generate a subsethood measure on $\overline{\mathcal{RF}}(\mathcal{U})$. To this end, we first prove the following proposition:

Proposition 3. *Let $\nu : \overline{\mathcal{RF}}(\mathcal{U}) \to [0,1]$ be defined as*

$$\nu(A) = \frac{1}{w-v}\int_v^w A(u)du, \tag{8}$$

in which the integral is the Lebesgue integral. This function is increasing. Moreover ν satisfies $\nu(\emptyset) = 0$ and $\nu(\mathcal{U}) = 1$.

Proof. Let $\sqsubseteq$ be the partial order in $\overline{\mathcal{RF}}(\mathcal{U})$ induced by $\sqcup$ and $A, B \in \overline{\mathcal{RF}}(\mathcal{U})$ such that $A \sqsubseteq B$. Equivalently, we have $A \sqcup B = B$. In the first case of Equation 7, every $u \in [\underline{a}^1 \wedge \underline{b}^1, \overline{a}^1 \vee \overline{b}^1]$ satisfies $A(u) \leq 1 = (A \sqcup B)(u) = B(u)$. In the second case, $A(u) \leq A(u) \vee B(u) = (A \sqcup B)(u) = B(u)$. We conclude that $A \sqsubseteq B$ implies $A \subseteq B$. Due to the properties of the Lebesgue integral, ν is increasing. Evidently, $\nu(\emptyset) = 0$ and $\nu(\mathcal{U}) = 1$. □

Corollary 1. *Let I be a fuzzy implication that satisfies $I(a,b) = 1$ for all $a \leq b \in [0,1]$ and ν be as in Equation 8. The operator $S^{\sqcup}_{I,\nu} : \overline{\mathcal{RF}}(\mathcal{U})^2 \to [0,1]$ given by*

$$S^{\sqcup}_{I,\nu}(A,B) = I(\nu(A \sqcup B), \nu(B)), \tag{9}$$

is a subsethood measure on $\overline{\mathcal{RF}}(\mathcal{U})$.

Recall that the residual implication I_T of a t-norm T satisfies $I_T(a,b) = 1$ for all $a \leq b \in [0,1]$. In addition, the following corollary holds true:

Proposition 4. *Let I_T be the residual implication of a t-norm T and $\nu : \overline{\mathcal{RF}}(\mathcal{U}) \to [0,1]$ be increasing with $\nu(\emptyset) = 0$ and $\nu(\mathcal{U}) = 1$. If $B \in \overline{\mathcal{RF}}(\mathcal{U})$ is such that $\nu(B) > 0$, then $S^{\sqcup}_{I_T,\nu}(A,B) > 0$ for any $A \in \overline{\mathcal{RF}}(\mathcal{U})$.*

Proof. Since T is a t-norm, $T(a,b) \leq T(1,b) = b$ for all $a, b \in [0,1]$. Hence, we have for all $a, b \in [0,1]$ that $b \in \{z \in [0,1] \,|\, T(a,z) \leq b\}$, which implies $b \leq \bigvee\{z \in [0,1] \,|\, T(a,z) \leq b\} = I_T(a,b)$. Thus, $0 < \nu(B) \leq I_T(\nu(A \sqcup B), \nu(B)) = S^{\sqcup}_{I_T,\nu}(A,B)$. □

This proposition can in particular be applied to the function ν of Equation 8 and we obtain the following corollary:

Corollary 2. *Let I_T be the residual implication of a t-norm T and ν be as in Equation 8. If $\int_v^w B(u)du > 0$, then $S^{\sqcup}_{I_T,\nu}(A,B) > 0$ for every $A \in \overline{\mathcal{RF}}(\mathcal{U})$.*

Equation 9 can be used to define a new subsethood measure on $\overline{\mathcal{RF}}(\mathcal{U})$. We begin by introducing some notations that will be used throughout this section. Let $\mathcal{U} = [\underline{u}, \overline{u}] \subset \mathbb{R}$ be a non-empty interval, $A \in \mathcal{RF}(\mathcal{U})$, and $\mathcal{A} = \{A_j\}_{j=1,\ldots,r} \subset \mathcal{RF}(\mathcal{U})$ such that $A_i \not\subseteq A_j$ for $i \neq j$. For convenience, we will refer to the elements of $\mathcal{A}$ as antecedents. Let the symbols $\mathcal{C} = [\underline{c}, \overline{c}]$ and $\mathcal{C}_j = [\underline{c}_j, \overline{c}_j]$ denote the cores of the input A and the antecedents $A_j \in \mathcal{A}$, respectively. For the rest of the article, we will assume that $\underline{c}_1 \leq \overline{c}_1 < \underline{c}_2 \leq \overline{c}_2 < \ldots < \underline{c}_r \leq \overline{c}_r$, indicating that the antecedents are ordered from left to right in terms of the relative positions of their cores. This *left to right* ordering of antecedents in terms of their cores enables to define the relative position of the input A with respect to each antecedent A_j, for all $j \in \{1, \ldots, r\}$. The idea of analyzing the relative positions between fuzzy sets has previously been employed to order and rank fuzzy sets [28].

In an upcoming journal article, we will define a subsethood measure $S^{\mathcal{A}}$ on $\overline{\mathcal{RF}}(\mathcal{U})$ such that $S^{\mathcal{A}}(A, A_j) > 0$ for $A \neq \emptyset$ if and only if one of the following conditions is fulfilled:

C1) $\mathcal{C} \cap \mathcal{C}_j \neq \emptyset$;
C2) $\overline{c}_j < \underline{c} < \overline{c} < \underline{c}_{j+1}$, for $j \in \{1, \ldots, r-1\}$;
C3) $\overline{c}_{j-1} < \underline{c} < \overline{c} < \underline{c}_j$, for $j \in \{2, \ldots, r\}$;
C4) $\overline{c} < \underline{c}_1$;
C5) $\overline{c}_r < \underline{c}$.

Conditions C1 - C5 formalize the idea that rule activation by means of $S^{\mathcal{A}}(A, A_j)$ depends on the relative position of the input's core $\mathcal{C}$ in comparison to the antecedent's core $\mathcal{C}_j$. In particular, C1 ensures that $S^{\mathcal{A}}(A, A_j)$ returns non-zero values for antecedents A_j whose cores are intersected by $\mathcal{C}$. Otherwise, C2 and C3 imply that $S^{\mathcal{A}}(A, A_j) > 0$ if and only if $\mathcal{C}$ is adjacent to $\mathcal{C}_j$. Conditions C4 and C5 account for the extreme cases in which $\mathcal{C}$ lies to the left of $\mathcal{C}_1$, or to the right of $\mathcal{C}_r$.

Due to space limitations, we are only able to describe the basic properties of $S^{\mathcal{A}}$. This subsethood measure is based on $S^{\sqcup}_{I_T,\nu}$ that satisfies $S^{\sqcup}_{I_T,\nu}(A, A_j) > 0$ for every $A \in \mathcal{RF}(\mathcal{U})$ whenever $\int_v^w A_j(u)du > 0$ (cf., Corollary 2). For each j, the calculation of $S^{\mathcal{A}}(A, A_j)$ involves evaluating certain expressions $S^{L,\mathcal{A}}(A, A_j)$ and $S^{R,\mathcal{A}}(A, A_j)$, where the superscripts L and R stand for left and right, respectively. In particular, $S^{L,\mathcal{A}}(A, A_j)$ is given by

$$S^{L,\mathcal{A}}(A, A_j) = \kappa^L_j(A) \cdot S^{\sqcup}_{I_T,\nu}(A, A_j), \tag{10}$$

where $\kappa^L_j(A) \in [0,1]$ depends on the relative positions of $\mathcal{C}$ and $\mathcal{C}_j$ as well as the distance between them. Specifically, κ^L_j is a function $\mathcal{RF}(\mathcal{U}) \to [0,1]$ that is chosen so as to satisfy the following:

1) If $\mathcal{C} \cap \mathcal{C}_j \neq \emptyset$, then $\kappa^L_j(A) = 1$.
2) If $\mathcal{C} \cap \mathcal{C}_j = \emptyset$ and $\mathcal{C}$ satisfies Condition C3) or C4, then $\kappa^L_j(A) \in (0,1)$. The larger the distance between $\mathcal{C}$ and $\mathcal{C}_j$, the smaller the value $\kappa^L_j(A) \in (0,1)$.
3) $\kappa^L_j(A) = 0$, otherwise.

One can define $S^{R,\mathcal{A}}(A, A_j)$ in a similar way. With $S^{L,\mathcal{A}}(A, A_j)$ and $S^{R,\mathcal{A}}(A, A_j)$ at hand, the firing strength of the j-th rule is given by

$$S^{\mathcal{A}}(A, A_j) = S^{L,\mathcal{A}}(A, A_j) \vee S^{R,\mathcal{A}}(A, A_j), \tag{11}$$

ensuring that only antecedents whose cores are intersecting or immediately adjacent to that of the input, to the left or to the right, contribute to the activation process. Figure 1 provides a schematic illustration of the rules activated in a sparse rule base when using $S^{\mathcal{A}}$.

IV. A Strategy for Handling Sparse Rule Bases Without FRI

Consider the following set of k fuzzy rules:

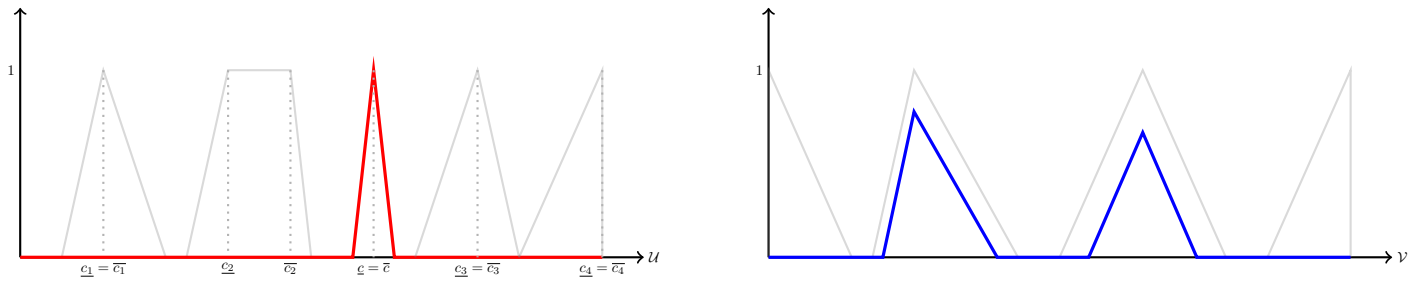

Fig. 1: Rule activation using the $\mathcal{A}$-FIS with a sparse rule base. Left: input A (red) and a set of antecedents $\mathcal{A} = \{A_i\}_{i\in\mathbb{N}_4}$, ordered from left to right. In this case, only $S^{\mathcal{A}}(A, A_2)$ and $S^{\mathcal{A}}(A, A_3)$ are nonzero. Right: the set of consequents $\mathcal{B} = \{B_i\}_{i\in\mathbb{N}_4}$ (gray), ordered from left to right, and the output of the $\mathcal{A}$-FIS (blue). Only consequents B_2 and B_3 are activated. The shape of the output depends on the operators used by the function Ψ.

$$R_i : \text{If } \boldsymbol{x} \text{ is } A_i \text{ then } y \text{ is } B_i, \text{ for all } i \in \{1, \ldots, k\} \quad (12)$$

As argued in [17], $\boldsymbol{x}$ *is* A_i can be interpreted as *the degree to which* $\boldsymbol{x}$ *is a subset of* A_i, with the rule being activated according to this degree. Given its capacity to detect the closest antecedents to the input and to activate rules even with no overlap between input and antecedents, the $S^{\mathcal{A}}$ subsethood measure can be used as a matching function in an ARS, defined as follows:

Definition 3. *Let* $\mathcal{U}_1, \ldots, \mathcal{U}_n, \mathcal{V}$ *be closed non-empty subintervals of* $\mathbb{R}$, $\mathbb{L} = \overline{\mathcal{RF}}(\mathcal{U}_1) \times \ldots \overline{\mathcal{RF}}(\mathcal{U}_n)$, $\mathcal{A} = \{A_i\}_{i\in\{1,\ldots,k\}} \subseteq \mathbb{L}$ *such that* $A_i \not\leq A_j$ *for all* $i \neq j$, *and* $\mathcal{B} = \{B_i\}_{i\in\{1,\ldots,k\}} \subseteq \mathcal{F}(\mathcal{V})$. *The* $\mathcal{A}$*-FIS given by a function* $\Psi : \mathbb{L} \to \mathcal{F}(\mathcal{V})$ *that computes the following fuzzy output* B *for an input* $A \in \mathbb{L}$*:*

$$B(v) = \Psi(A)(v) = \bigvee_{i\in\{1,\ldots,k\}} T(S^{\mathcal{A}}(A, A_i), B_i(v)), \quad (13)$$

for all $v \in \mathcal{V}$, *where* $T : [0,1]^2 \to [0,1]$ *is a t-norm.*

Figure 1 illustrates the rule activation process in the $\mathcal{A}$-FIS for a specific case of a sparse rule base. If the fuzzy implication I used to calculate $S^{\mathcal{A}}$ is parametric, we denote the dependence of the $\mathcal{A}$-FIS on its parameter as Ψ^{α}. According to Proposition 1, $S^{\mathcal{A}}$ can be constructed by aggregating subsethood measures corresponding to each dimension of the problem. Specifically, if $\mathcal{A}_i \subseteq \overline{\mathcal{RF}}(\mathcal{U}_i)$ represents the subset of fuzzy numbers characterizing the i-th dimension, $S^{\mathcal{A}}$ can be constructed by aggregating the n $\mathcal{A}_i$-subsethood measures associated with these subsets.

To effectively apply the $\mathcal{A}$-FIS for regression and time series prediction, we developed a strategy based on Algorithms 1, 2, and 3. First, each training point is associated with a subset of rules $\mathcal{R}_p$, selected based on two criteria: the proximity between its target value and the centroids of the consequents, and the value of the subsethood measure $S^{\mathcal{A}}$ computed between the input of the training point and the antecedents of the rules in $\mathcal{R}_p$, as detailed in Algorithm 1. Next, each training point p is associated with a inference system Ψ_p, optimized to ensure accurate predictions for that point. Finally, predictions for new inputs $\boldsymbol{x}$ are generated by aggregating the defuzzified outputs $\overline{\Psi_j(\boldsymbol{x})}$ of the set of subsystems $\{\Psi_j\}_{j\in\{1,\ldots,r\}}$ corresponding to the r closest training points to the input, weighted by their distances to the input.

Algorithm 1 Rule Base Assignment per Training Point. A^R and B^R denote the antecedents and consequent of rule R.

Input: Training dataset $\mathcal{X} = \{(\boldsymbol{x_p}, t_p) : p = 1, \ldots, m\} \subset \mathbb{R}^n \times \mathbb{R}$. Rule base $\mathcal{R} = \{R_i\}_{i\in\{1,\ldots,k\}} = \{(A^{R_i}, B^{R_i})\}_{i\in\{1,\ldots,k\}}$. Arbitrary parameter α, if $S^{\mathcal{A}}$ is parametric.
Output: Set of rules per training point $\{\mathcal{R}_p\}_{p\in\{1,\ldots,m\}}$.
Step 1: Compute the set $\mathcal{B}$ of unique consequents in the rule base, and $\gamma = |\mathcal{B}|$. Name its elements as $B_1, \ldots, B_\gamma$, according to the order of their centroids $c_1 \leq \ldots \leq c_\gamma$.
Step 2: Assign rules for each training point:
for $p \in \{1, \ldots, m\}$ **do**
  Identify $j < \gamma$ such that $t_p \in [c_j, c_{j+1})$ (or $j = 1$ if $t_p < c_1$, $j = \gamma$ if $t_p > c_\gamma$).
  $LR = \{R \in \mathcal{R} | B^R = B_j\}$.
  $RR = \{R \in \mathcal{R} | B^R = B_{j+1}\}$.
  $R_L = \underset{R\in LR}{\text{argmax}} S^{\mathcal{A}}(\boldsymbol{x_p}, A^R)$.
  $R_R = \underset{R\in RR}{\text{argmax}} S^{\mathcal{A}}(\boldsymbol{x_p}, A^R)$.
  $\mathcal{R}_p = \{R_L, R_R\}$ (or $\{R_L\}$ if $j = \gamma$, or $\{R_R\}$ if $j = 1$).
end for
return $\{\mathcal{R}_p\}_{p\in\{1,\ldots,m\}}$.

V. Experimental Application

Our primary objective is to apply the $\mathcal{A}$-FIS approach and compare its results against those reported by the FRI methods in [6] and [18]. The precision was evaluated using root mean squared error (RMSE) on seven benchmark problems obtained from [29]–[31], including three multivariate regression tasks and four time series prediction problems. We refer to the methods of interest as CC [18], T-FRI$_{\text{w}}$ [6], and $\mathcal{A}$-FIS.

A. Experimental Setup

Our setup follows the methodology described in detail in [6], [18]. One key difference is that for the $\mathcal{A}$-FIS approach,

Algorithm 2 Construction and Adjustment of Each $\mathcal{A}_p$-FIS.

Input: Training dataset $\mathcal{X}$, set of local rule bases $\{\mathcal{R}_p\}_{p\in\{1,\ldots,m\}}$.
Output: Set of inference systems $\{\Psi_p\}_{p\in\{1,\ldots,m\}}$.
for $p \in \{1,\ldots,m\}$ **do**
  Construct $\mathcal{A}_p$-FIS:
  $\mathcal{A}_p = \{A^R \mid R \in \mathcal{R}_p\}$, $S^{\mathcal{A}_p}$ from Eq. 11, and Ψ_p per Definition 3.
  if $S^{\mathcal{A}}$ depends on parameter α **then**
    Optimize α_p: $\alpha_p = \underset{\alpha}{\operatorname{argmin}} \|\overline{\Psi_p^{\alpha}(\boldsymbol{x_p})} - t_p\|_2$, and update $\Psi_p = \Psi_p^{\alpha_p}$.
  end if
end for
return $\{\Psi_p\}_{p\in\{1,\ldots,m\}}$.

Algorithm 3 Output Calculation.

Input: Input $\boldsymbol{x} \in \mathbb{R}^n$, distance measure d, number of closest points r to $\boldsymbol{x}$, training dataset $\mathcal{X}$, inference systems $\{\Psi_p\}_{p\in\{1,\ldots,m\}}$.
Output: Predicted value y.
Step 1: Normalize training points $\{\boldsymbol{x_p}\}$ and input $\boldsymbol{x}$ to $[0,1]$, obtaining $\{\tilde{\boldsymbol{x}_p}\}$ and $\tilde{\boldsymbol{x}}$.
Step 2: Identify the r closest points to $\tilde{\boldsymbol{x}}$, $\mathcal{CP} = \{\tilde{\boldsymbol{x}_j}\}_{j\in\mathcal{J}}$, where $\mathcal{J} \subseteq \{1,\ldots,m\}$ and $|\mathcal{J}| = r$.
Step 3: Compute the output:
$D = \sum_{j\in\mathcal{J}} d(\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{x}_j})$
$y = \sum_{j\in\mathcal{J}} \left(\left(1 - \frac{d(\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{x}_j})}{D}\right) \cdot \overline{\Psi_j(\boldsymbol{x})}\right)$
return y

instead of applying the standard Wang-Mendel (WM) rule generation method [32] by selecting the m rules with the highest strength, we selected the best m_i rules for each consequent B_i generated by the WM algorithm, to ensure a balanced representation of the output universe V. This selection was done such that the total number of rules did not exceed those used in [18]. The total amount of rules for each problem is detailed in Table I. Li et al. perform a fair comparison of their T-FRI$_{\text{w}}$ method with the CC method of Chen et al. using the datasets indicated in Table II. The former do not explicitly state the numbers of rules used by the T-FRI$_{\text{w}}$ method in these regression problems. However, no reduction in the numbers of rules is mentioned.

The subsethood measure $S^{\mathcal{A}}$ was constructed using the parametric Dombi fuzzy implication [33] and the Lukasiewicz t-norm was employed in Equation 13. To account for the varying relevance of features in each problem, subsethood measures were aggregated across dimensions using a weighted average, as justified by Proposition 1. The weights were derived from the Pearson correlation coefficients [34] between features and outputs, calculated on the training data. Specifically, each weight was defined as the absolute value of the correlation between the corresponding feature and the output. The parameter α for the Dombi implication required by Algorithm 2 was optimized using a grid search over $[0.1, 50]$ with a step size of 0.1. The number of closest points r was set to 10, and the L_1 norm was used as the distance d, while defuzzification was performed using the centroid method. The results shown in Table II are the averages over ten applications of five-fold cross-validation. All coding and computations were carried out using Python 3.10.

Dataset	CC [18]	$\mathcal{A}$-FIS
Abalone	39	24
Concrete Compressive Strength	36	35
Concrete Slump Test	25	11
Mackey Glass	not informed	40
Chemical Process Concentration	36	19
Chemical Process Temperature	21	17
Gas Furnace	66	55

TABLE I: Number of Generated Rules

B. Experimental Results

Table II presents the RMSE values obtained by the different methods, highlighting that the $\mathcal{A}$-FIS approach achieve the lowest error in three cases and ranks second in four. Table III presents the average normalized RMSE, where normalization is performed separately for each dataset by mapping its worst observed result to 1 and its best to 0. The results indicate that the $\mathcal{A}$-FIS approach achieves a lower average normalized RMSE compared to the reference methods, with an overall error reduction of 46.4% relative to the T-FRI$_{\text{w}}$ method.

Dataset	CC [18]	T-FRI$_{\text{w}}$ [6]	$\mathcal{A}$-FIS
Abalone	2.45	2.4639	2.3503
Concrete Compressive Strength	13.44	8.8001	9.4064
Concrete Slump Test	5.91	3.6576	4.2587
Mackey Glass	0.0597	0.0391	0.0374
Chemical Process Concentration	0.3248	0.3154	0.2914
Chemical Process Temperature	0.2241	0.3673	0.3304
Gas Furnace	0.7035	0.6214	0.6509

TABLE II: RMSE Comparison Across Datasets

	CC [18]	T-FRI$_{\text{w}}$ [6]	$\mathcal{A}$-FIS
Average	0.8397	0.3993	0.2142

TABLE III: Average Normalized RMSE Comparison

VI. Conclusion

This paper describes the basic features of an analogical FIS called $\mathcal{A}$-FIS that is based on a novel subsethood measure, denoted $S^{\mathcal{A}}$, which is defined on the bounded lattice of fuzzy numbers $\overline{\mathcal{RF}}(\mathcal{U})$. (Due to restrictions in the number of pages, a detailed description of the $\mathcal{A}$-FIS must be postponed to an upcoming journal article.) The $S^{\mathcal{A}}$ subsethood measure enables inference with sparse rule bases by directly identifying and activating rules whose antecedents are closer to the input without requiring FRI. We propose an algorithmic approach

for regression and time series prediction. For each training point $\boldsymbol{x_i}$, the method builds a specific $\mathcal{A}_i$-FIS model. The final output is obtained by averaging the defuzzified predictions of r $\mathcal{A}_i$-FIS models, corresponding to the r training points closest to the input $\boldsymbol{x}$. The average is weighted according to the distance between $\boldsymbol{x}$ and each $\boldsymbol{x_i}$, so that closer points have a stronger influence. The proposed method was evaluated on seven benchmark datasets and compared with two competitive fuzzy rule interpolation techniques. The $\mathcal{A}$-FIS achieved the lowest average normalized RMSE, demonstrating a 46.4% improvement over T-FRI$_{\text{w}}$ [6]. Notably, the $\mathcal{A}$-FIS required fewer rules than the referential FRI methods, showcasing its efficiency and effectiveness.

Despite these promising results, several areas warrant further investigation. While the $\mathcal{A}$-FIS achieved robust overall performance, our model did not consistently outperform other methods across most datasets. Moreover, selecting the optimal combination of t-norms, fuzzy implications, and other parameters introduces additional complexity during configuration and training. Future research will focus on developing efficient training techniques, conducting a comprehensive analysis of stability and computational complexity, and exploring strategies to enhance the method's performance further.

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